In [87]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

In [109…
```
img = cv2.imread("1.jpeg", cv2.IMREAD_COLOR)

# Split the image into the B,G,R components
b, g, r = cv2.split(img)

# Show the channels
plt.figure(figsize = (20, 5)) #SIZE DIMENSIONS

# Red Channel
plt.subplot(1, 4, 1)
plt.imshow(r, cmap = "gray")
plt.title("Red Channel")
plt.axis("off")

# Green Channel
plt.subplot(1, 4, 2)
plt.imshow(g, cmap = "gray")
plt.title("Green Channel")
plt.axis("off")

# Blue Channel
plt.subplot(1, 4, 3)
plt.imshow(b, cmap = "gray")
plt.title("Blue Channel")
plt.axis("off")

# Merge the individual channels into a BGR image
merged_img = cv2.merge((b, g, r))
rgb_img = cv2.cvtColor(merged_img, cv2.COLOR_BGR2RGB)

# Show the merged output
plt.subplot(1, 4, 4)
plt.imshow(rgb_img)
plt.title("Merged Output")
plt.axis("off")
plt.show()
```

In [104…
```
#DEFAULT => BGR
#CONVERT => COLOR_BRG2--- [HSV, GRAY, RGB]
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)
img_rgb
```

Out[104…

```
array([[[103,  49,  83],
        [103,  49,  83],
        [103,  50,  82],
        ...,
        [107,  57, 187],
        [110,  55, 187],
        [110,  55, 187]],

       [[106,  46,  83],
        [106,  46,  83],
        [103,  49,  83],
        ...,
        [108,  56, 187],
        [110,  55, 187],
        [110,  54, 188]],

       [[106,  34,  83],
        [111,  40,  83],
        [106,  45,  85],
        ...,
        [110,  55, 187],
        [110,  55, 187],
        [110,  55, 187]],

       ...,

       [[ 20, 106, 120],
        [ 20, 106, 120],
        [ 20, 106, 120],
        ...,
        [ 18, 149, 116],
        [ 18, 151, 115],
        [ 18, 152, 114]],

       [[ 20, 106, 120],
        [ 20, 106, 120],
        [ 20, 106, 120],
        ...,
        [ 18, 152, 114],
        [ 18, 152, 114],
        [ 18, 153, 113]],

       [[ 20, 106, 120],
        [ 20, 107, 119],
        [ 20, 107, 119],
        ...,
        [ 18, 152, 114],
        [ 18, 152, 114],
        [ 18, 153, 113]]], dtype=uint8)
```

In [107…

```
print("Shape of the image: ", img_gray.shape)
print("Data type of the image: ", img_gray.dtype)
print("Size of the image: ", img_gray.size)
```

```
Shape of the image:  (3060, 4080)
Data type of the image:  uint8
Size of the image:  12484800
```